

SONiX 32-Bit Cortex-M0 Micro-Controller

SPI Example Code

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AMENDMENT HISTORY

Version	Date	Description
1.0	2024/12/02	1. First version.

Table of Content

AMENDMENT HISTORY	2
1 OVERVIEW.....	4
2 FEATURE	4
3 HW	4
4 EXAMPLE CODE.....	5
4.1 POLLING FLAGS.....	5
4.2 INTERRUPT.....	5
5 RESULT	7
5.1 POLLING FLAGS.....	7
5.2 INTERRUPT.....	8

1 OVERVIEW

This document provides our customers the C program examples of SPI.

2 FEATURE

- Master & slave mode
- Transmitter & receiver mode
- Interrupt or polling mode
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.
- Connect MOSI0 to MISO0.

4 EXAMPLE CODE

4.1 Polling Flags

1. Set SPI0 to transmit 32 bytes.

```

26  /*_____ D E C L A R A T I O N S _____*/
27  void PFPA_Init (void);
28
29  /*_____ D E F I N I T I O N S _____*/
30  #define SPI_TEST_CNT 32
31
32  uint16_t wSPI_Tx_Fifo[SPI_TEST_CNT];
33  uint16_t wSPI_Rx_Fifo[SPI_TEST_CNT];
34

```

2. Call SPI0_Init ();

```

int main (void)
{
    //initSPI0
    SPI0_Init ();

    .
    .
    User code
    .
    .
}

```

4.2 Interrupt

1. Set SPI0 to transmit 32 bytes.

```

26  /*_____ D E C L A R A T I O N S _____*/
27  void PFPA_Init (void);
28
29  /*_____ D E F I N I T I O N S _____*/
30  #define SPI_TEST_CNT 32
31
32  uint16_t wSPI_Tx_Fifo[SPI_TEST_CNT];
33  uint16_t wSPI_Rx_Fifo[SPI_TEST_CNT];
34

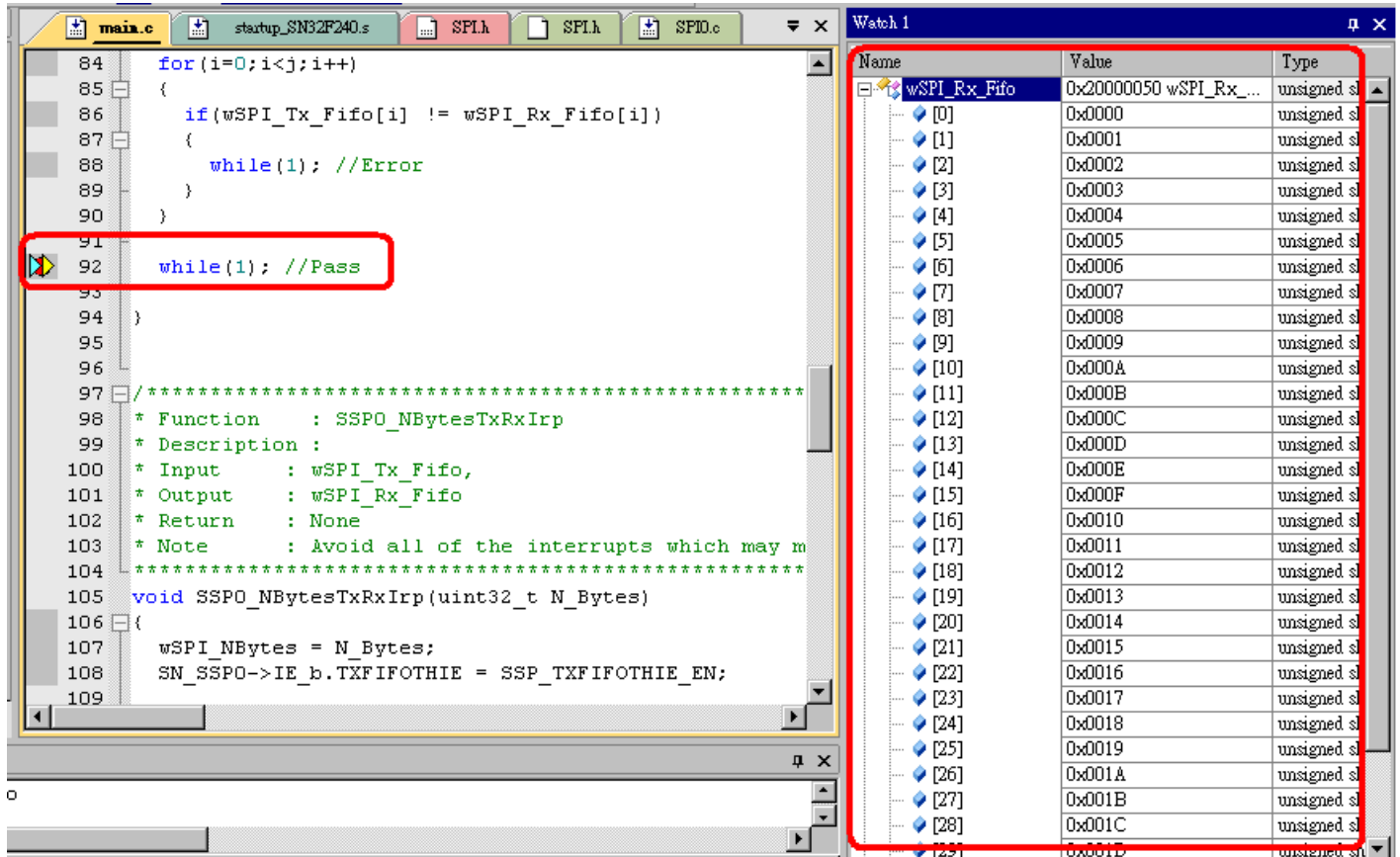
```

```
2. Call SPI0_Init ();  
int main (void)  
{  
    //init SPI0  
    SPI0_Init ();  
  
    .  
    .  
    User code  
    .  
    .  
}
```

5 RESULT

5.1 Polling Flags

The program will stop at the break point in main loop. The wSPI_Rx_Fifo will show correct data in watch memory window.

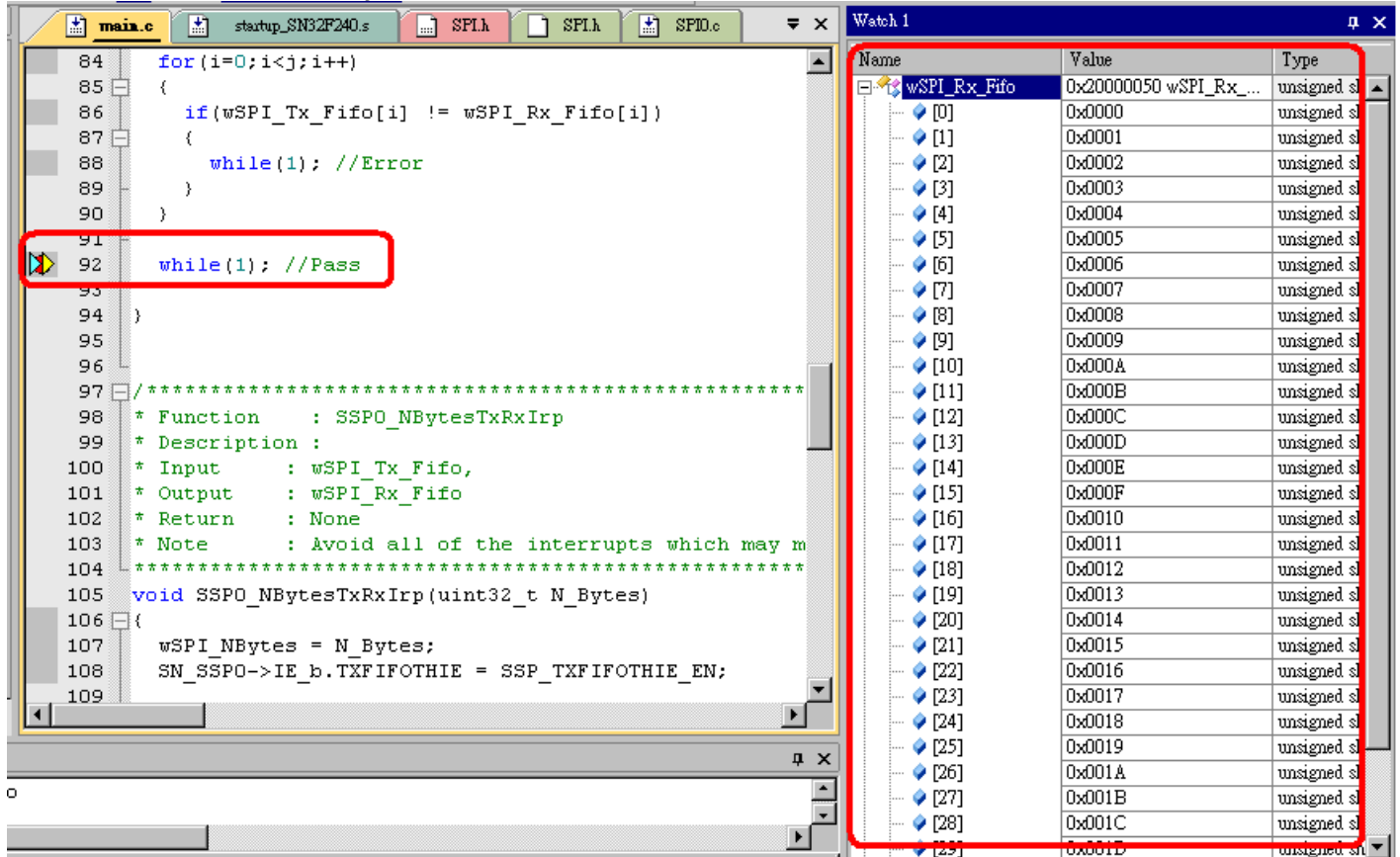


The screenshot shows a debugger interface with two main windows. The left window displays the source code of `main.c`. A red rectangle highlights the line `while(1); //Pass` at index 92, which is the current execution point. The right window, titled "Watch 1", displays the contents of the `wSPI_Rx_Fifo` array. The array is of type `unsigned short` and contains 16 elements, indexed from 0 to 15. The values are sequential, starting from `0x0000` at index 0 and ending with `0x000F` at index 15. The array is named `wSPI_Rx_Fifo` and its base address is `0x20000050`.

Name	Value	Type
wSPI_Rx_Fifo	0x20000050 wSPI_Rx_...	unsigned short
[0]	0x0000	unsigned short
[1]	0x0001	unsigned short
[2]	0x0002	unsigned short
[3]	0x0003	unsigned short
[4]	0x0004	unsigned short
[5]	0x0005	unsigned short
[6]	0x0006	unsigned short
[7]	0x0007	unsigned short
[8]	0x0008	unsigned short
[9]	0x0009	unsigned short
[10]	0x000A	unsigned short
[11]	0x000B	unsigned short
[12]	0x000C	unsigned short
[13]	0x000D	unsigned short
[14]	0x000E	unsigned short
[15]	0x000F	unsigned short

5.2 Interrupt

The program will stop at the break point in main loop. The wSPI_Rx_Fifo will show correct data in watch memory window.



The screenshot shows a debugger interface with the `main.c` file open. A break point is set at line 92, which is highlighted with a red box. The code at line 92 is `while(1); //Pass`. The Watch 1 window on the right shows the `wSPI_Rx_Fifo` array, which is also highlighted with a red box. The array contains 29 elements, each of type `unsigned short`, with values ranging from `0x0000` to `0x001D`.

Name	Value	Type
wSPI_Rx_Fifo	0x20000050 wSPI_Rx_...	unsigned st...
[0]	0x0000	unsigned st...
[1]	0x0001	unsigned st...
[2]	0x0002	unsigned st...
[3]	0x0003	unsigned st...
[4]	0x0004	unsigned st...
[5]	0x0005	unsigned st...
[6]	0x0006	unsigned st...
[7]	0x0007	unsigned st...
[8]	0x0008	unsigned st...
[9]	0x0009	unsigned st...
[10]	0x000A	unsigned st...
[11]	0x000B	unsigned st...
[12]	0x000C	unsigned st...
[13]	0x000D	unsigned st...
[14]	0x000E	unsigned st...
[15]	0x000F	unsigned st...
[16]	0x0010	unsigned st...
[17]	0x0011	unsigned st...
[18]	0x0012	unsigned st...
[19]	0x0013	unsigned st...
[20]	0x0014	unsigned st...
[21]	0x0015	unsigned st...
[22]	0x0016	unsigned st...
[23]	0x0017	unsigned st...
[24]	0x0018	unsigned st...
[25]	0x0019	unsigned st...
[26]	0x001A	unsigned st...
[27]	0x001B	unsigned st...
[28]	0x001C	unsigned st...
[29]	0x001D	unsigned st...

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Main Office:

Address: 10F-1, NO. 36, Taiyuan Street, Chupei City, Hsinchu, Taiwan R.O.C.
Tel: 886-3-5600 888
Fax: 886-3-5600 889

Taipei Office:

Address: 15F-2, NO. 171, Song Ted Road, Taipei, Taiwan R.O.C.
Tel: 886-2-2759 1980
Fax: 886-2-2759 8180

Hong Kong Office:

Unit No.705, Level 7 Tower 1, Grand Central Plaza 138 Shatin Rural Committee Road, Shatin, New Territories, Hong Kong.
Tel: 852-2723-8086
Fax: 852-2723-9179

Technical Support by Email:

Sn8fae@sonix.com.tw